



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-26)
Faculty Name	Ms. P. Swethanagasri
Student Name	SURABHI ANUSHKA
Roll Number	2520090168
Branch	CSIT
Course Outcome	3

Case Study Topic: Bangalore Metro Phase-3 expansion — minimum-cost connection plan with redundancy.

Work Done Summary:

Designed a minimum-cost metro expansion network connecting all six new stations (K, W, S, E, Y, H) to the central hub M (Majestic) while satisfying the regulatory requirement of at least two edge-disjoint paths between M and W (Whitefield). The solution first constructs a Minimum Spanning Tree (MST) to minimize total construction cost and then adds the lowest-cost additional edge necessary to create a second independent path between Majestic and Whitefield, ensuring network connectivity, fault tolerance, and compliance with redundancy regulations.

Implementation Details :

- **Model the metro network as a weighted undirected graph.**
- Represent stations as vertices and candidate rail links as weighted edges.
- Use **Kruskal's Algorithm** with a **Disjoint Set Union (DSU)** structure to build the MST.
- Verify the existence of two edge-disjoint paths between **M** and **W**.
- If the MST provides only one path, add the minimum-cost non-MST edge that creates an alternate route.
- The final graph remains connected and satisfies the redundancy mandate.

Result : Screenshots/Output

Selected Metro Links:

W -- E : ₹4 cr

K -- S : ₹5 cr

S -- E : ₹6 cr

K -- W : ₹7 cr

M -- K : ₹8 cr

W -- Y : ₹8 cr

Y -- H : ₹9 cr

Minimum Construction Cost = ₹47 crore

Time Complexity:

1. Sorting edges: $O(E \log E)$
2. DSU operations in Kruskal's Algorithm: $O(E \alpha(V))$
3. Redundancy verification and extra-edge insertion: $O(E)$
4. Overall Time Complexity: $O(E \log E)$

GitHub Link :

Student Signature	Faculty Signature